

Ljubomir JOSIFOVSKI

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Summary

ML/AI researcher and engineer in industrial R&D, with background spanning speech recognition, quantitative research, trading and machine learning systems. Experience building research platforms and production systems across forecasting, optimisation, and large-scale data-driven modelling.

Now: designing closed-loop LLM and agentic systems in which tools gather or act on evidence, independent evaluators inspect outputs, and structured traces, regression corpora and outcome ledgers make improvement measurable rather than anecdotal. Applying ideas from structured search and hypothesis management developed in ASR lattice decoding to train-time and test-time reasoning; interested in programmatic prompting (DSPy) and conservative, evidence-backed improvement of agent harnesses.

Prior: quantitative researcher, trader and portfolio manager developing systematic equity and FX models including low-SNR signal extraction, fitting, forecasting, portfolio optimisation, and risk management.

Earlier: PhD research in robust speech recognition in noise; MSc in text-to-speech synthesis and spoken document retrieval. Background in applied mathematics, statistics, computer science, and industrial R&D.

Skills

C/C++/OpenMP, MATLAB, bash, vim, awk, SQL, duckdb, PostgreSQL, MS SQL Server, c/make, gcc, gdb, ddd, shell tools, ssh, rsync, screen, VSCode, python, jupyter, Spyder, git, mercurial, cvs, MS Teams, github, R, Java, C#, Visual Studio, Slurm, Condor, Compute Cloud, Bloomberg terminal/API, Reuters Kobra, assembly.

Agents and evaluation: pi, Hermes, Codex, OpenCode, OMP, Cursor, Claude Code, Cline and local LLMs served through llama-server / hipfire / LM Studio; multi-model generation, tool execution, judge feedback, structured-output verification, regression evaluation, DSPy and GEPA.

Platforms

Linux (X/Ubuntu, CentOS), MacOS, MS-Windows (from DOS to v11), Cloud/cluster, Unix

Recent ML/AI systems and agents OSS work

- **LoveWork** — personal life work-discovery agent; thesis 1-most people underachieve relative to potential b/c chasing work/projects/interest leads is time-costly, and 2-would benefit from personal agent/guide through, democratizing human agents (now available only to movie&literary stars); LLM-guided crawling across sources of leads, candidate-aware scoring and lifecycle tracking, dashboard/MCP access, append-only decisions and outcome ledgers. Building a historic dataset with conservative reflection and evaluation harness, that proposes evidence-backed, regression-tested improvements when changing policy, backed by a probabilistic model of the process.
 - **ChEMBLdb-query** — LLM-assisted SQL over ChEMBL, including local-model endpoint meshes. Version 5 uses specialised generation and judge roles in bounded SQL → execution → feedback → revision loops, backed by a reproducible 1,010-case evaluation corpus and synthetic-data work; both data set curation and syntehtic data generation are designed into the forward-backward passes.
 - **Hermes Agent /advisor** — in-review persistent second-model review plugin: a separate model reviews the main agent's work each turn, preserves unresolved concerns by session, and returns structured inline advice without blocking the primary agent. Iterated in response to upstream PR feedback.
 - **Local/open-weight inference and reasoning:** extended the C/Metal [ds4 REAP compact-support branch](#) to run compact DeepSeek V4 Flash REAP GGUFs on Apple Silicon; [hipfire](#) branch work adds grammar-constrained generation, loop guards and separate reasoning budgets; ported experimental [llama.cpp](#) Metal support for [Ling-2.6-Flash](#).
 - **Model portability and reproducible systems:** [Nemotron 3 Elastic MLX conversion tools](#), [ROCm/local-LLM benchmarking](#), [Telegram voice + text customer-support](#).
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Work Experience

F9 Research, Harpenden, UK

May 2016 – Now

Position: Director

Quant research, development and trading. Portfolio manager, run a small market neutral book ~350M USD gross, trading ~35M USD daily in the EU markets (and a small R&D US book). Consulting for quant R&D for a client, working on higher frequencies and short horizons (seconds and minutes) in C/C++, OMP, python, Matlab, PostgreSQL, cloud boxes and Slurm cluster. Input into varying aspects of the R&D pipeline – from informing and assessing latest technologies (including ML) to interviewing new teams members.

Re-engaged with ML/AI via llama.cpp, open-weight local models, coding agents and LLM APIs; built local-agent workflows for Python, JavaScript, CSS/HTML and C++ debugging. Modelled transcript data with doc2vec and applied newer ML methods to tabular-data forecasting (cf. Hugging Face TabArena). Separately, the public work above develops agent evaluation/refinement loops, local inference and research tooling. F9 owns the IP to all F9 R&D.

FutureSearch, remote distributed (US, UK, EU)

October 2025 – December 2025

Position: Research Scientist

Startup working on AI agentic tools. Built and used agents that assess a research need, write code, gather and organise financial data, and iterate with internal curation and quality control for end-user product concepts and internal workflows. Consulted on evaluation of agent-generated data and signals for potential investment.

Marshall Wace, London, UK

February 2010 – March 2016

Position: Quantitative Researcher

On the TOPS QR team, senior team member among a handful of people, creating research, developing code, shepherding the market neutral portfolio growth from a few hundred millions to double digit billions USD gross book size. Ushered the idea of a single unified framework for all quant R&D&trading with standardised components – data ingestion and caching, signals extraction, modeller for forecasting, portfolio optimizer, trades simulator, standardised reporting, a baseline sim faithful and realistic to be continuously improved on by the entire team working in unison on various components of the system.

Wrote or significantly contributed to major components of the system through their iterative improvements over the years. Big projects in production improving the then best baseline: dynamic modeller fitting the alpha signals expected returns at multiple horizons, incorporating both prior knowledge, constraints, and the evidence from historical data, market impact model in the simulator and the optimizer including slippage monitoring tuning and balancing risk cost of undercharging with the opportunity cost of overcharging, liquid concentrated low TO high capacity market neutral portfolios, 150/50 portfolios mix of tracker and market neutral, shepherding the trade scheduler deployment in production, alphas signals GeoSales, Suppliers-Customers, Directors deals, various reverting signals, 1st quantitative research and assessment on the in-house Alpha Capture signal.

Guided and helped younger hires from onboarding to them becoming fully productive wholly effective team members. Pioneered reproducible research at scale using multi cpu multi core R&D boxes with establishing and popularising best practices.

Credit Suisse, London, UK

November 2007 – November 2009

Position: Quantitative Analyst

On the Index Arbitrage proprietary trading desk. Independently traded equity market/sector/factor neutral portfolios on multiple European markets, fully automated and systematic, non-discretionary. Wrote own trading, analytics, backtest and portfolio construction systematic trading platform consisting of a Matlab core, Mosek optimiser, bash/awk scripts, Reuters Kobra Excel and Sql for historic and current data, with integrated risk monitoring and control using Barra's style factors and sectors. Used the platform to research and trade all the strategies and portfolios. Alone did orders

generation, portfolio construction, forecasting & modelling, all data feeds (Reuters, Sql dumps), the daily monitoring, trading analysis and slippage tracking and any other R&D&ops as needed for trading.

Traded multiple portfolios daily of ~500 names in total on London, Paris, Frankfurt, Switzerland, Milan and Madrid exchanges, one trade per day per name. Did R&D simulations for intra-day horizons faster TO.

In 2008 traded the London portfolio most of the year as a test bed for all research & development, returning 10% gross in 230 days with Sharpe of 2.5. In 2009 traded bigger book on most of the European markets, returned 8% gross to Aug'09 with Sharpe of 5.2, turnover 2-3 days, one trade per name per day. All together lifetime (388 days) return on gross 18% at Sharpe of 3.1.

G-Research (part of the DPFM group), London, UK

July 2004 – September 2007

Position: Quantitative Analyst

Research (70%), development (20%), daily portfolio monitoring and support (10%) in a multi-billion market neutral hedge fund systematically trading global equities and spot FX round the clock in a completely automated system. Research and creation of new trading models/alphas, coding, testing in simulation and putting them in production. Models for volume prediction, fundamentals and technical equities models (multiple markets), spot FX – all productionised and live traded. Built futures models but not traded live. Development included coding up the models, the associated data analytics, and subsequent performance and integrity monitoring once live. The portfolio support role involved monitoring the trade flow, market conditions and risk factors, investigating/tuning the trading.

In the process familiarised myself with forecasting and modelling, performance attribution, multiperiod quadratic portfolio optimisation, risk measurement and management (Barra, APT, custom factors), real-time and historic data feeds, data aggregation. Independently came up with original alphas building on well known semi-parametric models for forecasting that were traded live in equities and spot FX trading. Similarly contributed alphas based on novel non-parametric models used for trading equities. They were all profitable, contributed to the bottom line and were traded along the other alphas.

Canon Research Europe, Bracknell, UK

June 2001 – June 2004

Position: Researcher

Research & development work in the Machine listening group on ASR and indexing & retrieval of spoken documents. Contributed to all aspects of Canon's low resource embedded multiplatform ASR engine: the front-end (DSP related), decoder (Mpeg7 compatible lattice creation), training & using statistical models (acoustic HMM multilingual, text-to-phone Ngrams). Group demonstrated embedded speaker independent phone book name dialling on ARM9 & ARM7 phones. Phonetic indexing of spoken documents/annotations & retrieval with spoken & written queries.

Invented & implemented in the embedded C++/C codebase novel algorithm for searching annotation (speech) lattices with a query (speech) lattice, outperforming other known techniques for phonetic SDR (LATTICE MATCHING, UK Patent App No 0316669.1, accomp app ref 2865001, Jul 2003). Demoed playlist entry selection by voice for an MP3 player, performing in near realtime on Windows CE platform with 1500 entries.

Motorola European Research Lab, Basingstoke, UK

November 2000 – June 2001

Position: Research engineer

Technology transfer from my PhD work to Motorola (my industrial sponsor). Research on the distributed speech recognition (DSR) ETSI Aurora 2 standard platform. Developed robust ASR algorithms in Matlab, GNU C/C++ and tested them on Cygwin, HP-UX and Linux platforms. Lab was part of the winning consortium of the ETSI Aurora 2 standardisation competition for mobile phones robust front-end.

Macedonian Banking Operations Centre (USAID funded project for technical support of the financial sector in Macedonia), Skopje, MK

November 1997 – January 1998

Position: Management Information Systems – Electronic Data Processing (MIS-EDP) Advisor

In a team of advisers analysing operations of commercial banks in Macedonia. Handled the MIS-EDP operations of the banks surveyed, reported on the state of and recommended improvements. By the end of the project all commercial banks in Macedonia volunteered to have their operations surveyed and reported on.

Faculty of Mechanical Engineering, University Sv. Kiril i Metodij, Skopje, MK

November 1993 – October 1997

Position: Systems engineer

Solely responsible for maintaining all faculty computers (100+ PCs, 10+ Unix workstations), faculty LAN spanning 3 buildings, other computing-related equipment (printers, terminal servers, router). Faculty LAN massively expanded, doubled the size of existing and added a second computerised classroom for students and lab classes, introduced email & other Internet services to every staff member and student, phased out legacy systems (VT420 terminals, terminal servers). Maintained/supported collection of legacy Clipper/FoxPro accounting applications.

NeoCom, Skopje, MK

June 1993 – October 1993

Position: System integrator

In small & dynamic company, clients facing, computer systems assembly, integration, software installation, maintenance (PC/Windows), computer networks (Novell NetWare, Windows LAN) installation & maintenance on- and off-site.

Freelance S/W developer, undergraduate & hobby programming

1986 – 1993

Basic & assembler (6502) on home computers. Mission critical (firing heavy guns) on pocket computers (HP-71B, Sharp 1500) and TurboPascal (Apple II+CP/M board+HDD) while national service (army). MS-DOS systems programming (C & assembler, TSR programs: screen capture, serial port snoop, DOS trashcan), network programming (NetBIOS based LAN messenger, IPX chat, IPX stack emulator in DesqView), PC databases (video shop rental application in Clipper, various applications in FoxPro, document flow in MS-Access).

Education

Doctor of Philosophy Ph.D. (Full-Time)

1998 – 2000

Speech and Hearing Group, Department of Computer Science, Faculty of Engineering, University of Sheffield, UK.

Independent research into recognising speech in noise. Missing data model treats parts of the speech spectrum swamped by noise as unobserved/partially observed, giving rise to a probabilistically modelled mask that has to be incorporated in the frame-by-frame adapted speech model. Work involved theory of automatic speech recognition as well as practice, training HMMs with continuous GMM pdfs using EM (HTK, shell scripting), writing and using Viterbi decoders and frontends to test novel noise robustness algorithms, noise and SNR estimation (Matlab, C++, C). Part of EC ESPRIT LTR programme funded RESPITE project of 5 research labs and 2 industrial partners and EC TMR programme funded SPHEAR network.

Thesis: “Robust speech recognition with missing and unreliable data”. (Viva Dec 2002)

M.Phil. Electrical Engineering (Part-Time)

1993 – 1997

Department for Computers and Informatics, Faculty of Electrical Engineering, University Sv. Kiril i Metodij, Skopje, MK.

Studies consisted of taught part (2 years/4 semesters) for 6 courses and thesis work (1 year/2 semesters) followed by a viva. Courses achieved avg grade 10 (scale 6-10, 10 best). Projects: video-over-IP frame rate control and QoS using UDP non-blocking sockets (C/C++, part of a system for tele-teaching system); database of Medieval Manuscripts (Delphi). Thesis/research – built system for converting written text into speech. Rudimentary time-domain, syllable based TTS. Created a database of 1200 syllables, wrote TTS engine breaking the input text into syllables (using an NN MLP), concatenating the units from the syllable database, generating F0 and the duration contours, modifying the syllable units accordingly in time domain. Gathered and labelled data, trained a two layer, feed forward MLP (neural network) to mark syllable breaks in the input text. Part of a larger project for automatic text reading for the blind.

Thesis: “System for text-to-speech conversion for Macedonian language”.

B.S. Electrical Engineering (Full-Time)

1988 – 1993

Department for Computers, Informatics and Automation, Faculty of Electrical Engineering, University Sv. Kiril i Metodij, Skopje, MK.

Taught studies 4.5 years (9 semesters) followed by a diploma work (1 semester) and public presentation. Achieved average grade of 8.78 (scale 6-10, 10 best).

Diploma project: “Introduction to DECNET, Bitnet (EARN) and Internet networks; E-mail/File transfer services; X.25 Network and out-dial NUAs”.

Best student within my college class in years 1 & 2. Ranked 1st (100 points out of 100) among of approx 800 candidates at the University entrance exams.

R.J. Korcagin High School, Skopje, MK

1983 – 1987

Mathematics and Computer Science High School, achieved GPA 5.00 on a 2 to 5 scale (5 best), voted best pupil (“valedictorian”) of the 1983-87 generation.

Personal

Nationality: UK (acquired by choice), Macedonian (got by birth). Born 1968.

Languages: English, Macedonian (native), Croatian, Serbian.

Honours & Awards: - Scholarships: merit research & science 1988-93, talented student 1983-87. Best student 1989, '90. - Maths competitions prizes: Regional 1st 1984, '86, '87, 3rd 1985; Republic 3rd 1984, '85, '87; National participation 1984, '85, praise 1987.

Other: UK and MK driving licences, married, two grown up children.

Interests: science, technology, innovation, knowledge, epistemology, culture, arts, non-fiction, systems theories, solar punk, political economy, quantitative finance, history, ethics, mentoring, e/acc.